RiskPulse

AI/NLP Risk Engine + Event-Driven Portfolio Stress Testing

Problem
Turn unstructured financial news into structured risk signals

Solution
Frozen NLP inference -> impact calibration -> portfolio stress

1. Problem & Approach

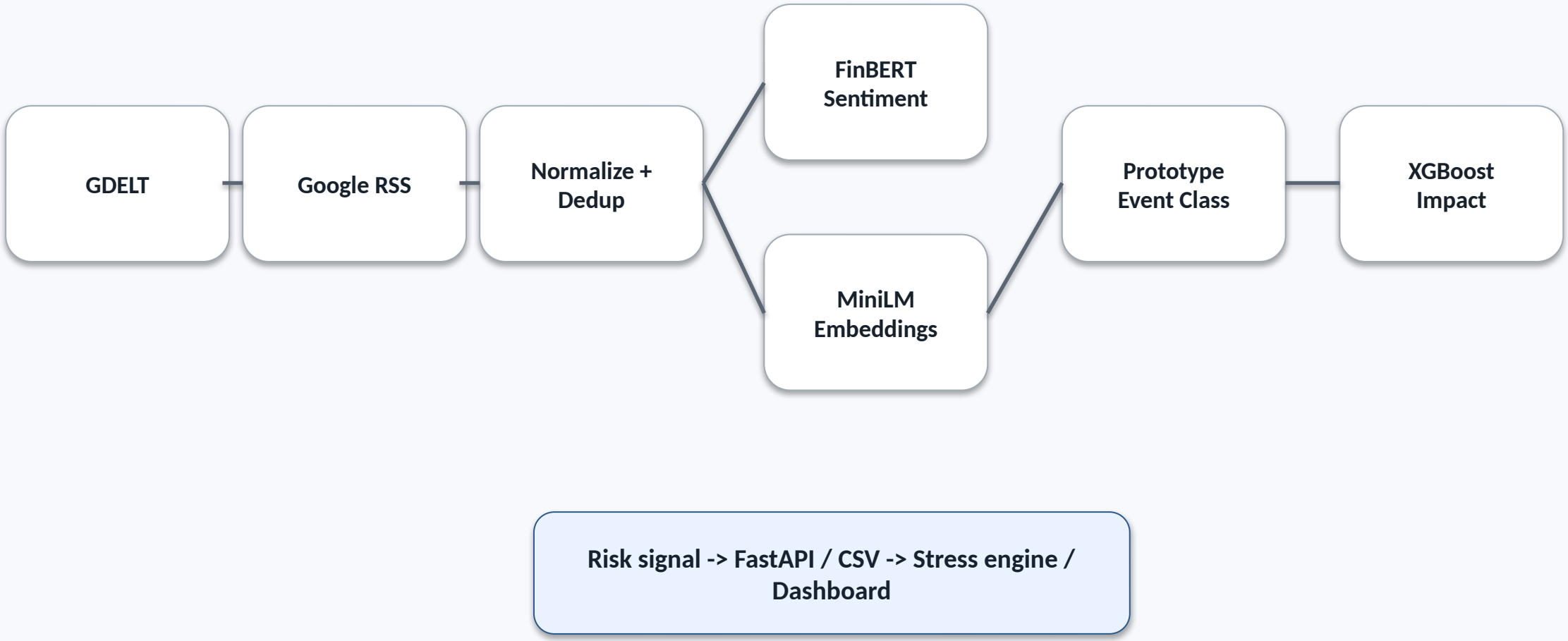
Ingest at least two public text sources and normalize them into one event stream.

Return sentiment (-1 to +1), event class, confidence, impact (1-10), entity and source metadata.

Choose Module B: event-driven stress testing. High-impact events (>=7) trigger transparent asset-level shocks.

Use frozen NLP models; train CPU XGBoost on public EDT next-day stock-move labels.

2. System Design



3. Implementation Highlights

GDELT DOC 2.x + Google News RSS are no-key public connectors; NewsAPI is optional through an environment variable.

FinBERT is domain-specific sentiment inference. MiniLM creates frozen sentence embeddings; event labels are selected by cosine similarity against fixed prototypes.

XGBoost regresses log absolute next-day stock moves from frozen 384-dimension MiniLM embeddings, trained on public EDT labels.

The stress engine maps event classes to explicit equity/rate/spread/PD shocks, then reports asset-level and portfolio-level loss.

4. Demo Outputs & Results

Structured signal
Sentiment + Event + Impact

Trigger
Impact >= 7

Stress test
Portfolio before / after

The included offline demo contains six representative financial events across macro, geopolitical, credit, cyber, earnings and M&A categories.

Model check: 50,000 rows; 10,000 chronological holdout. XGBoost MAE 1.699 vs 1.731 percentage points for a median baseline; improvement is small.

5. Domain Impact & Limitations

Business value: reduce time from news arrival to a machine-readable risk flag and a portfolio stress view.

Traceability: each portfolio loss can be decomposed into beta, duration, spread duration and credit-loss components.

Limitation: EDT labels target next-day single-stock movement, not portfolio loss. Some ticker assignments are wrong; the 50,000-row holdout MAE only narrowly beats the median baseline.

Next: strengthen issuer labels and test across more years and issuers.